Memory Notes

Simplified DDR4 Simplified edition - JEDEC JESD79-4D

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1 Simplified Ddr4 Index

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2 Introduction

2.1 What DDR4 SDRAM is

DDR4 SDRAM is the fourth generation of double-data-rate synchronous DRAM, standardized by JEDEC as JESD79-4D. It keeps the 8n prefetch of DDR3 but splits banks into groups, lowers the nominal supply to 1.2 V (SSTL_12), and adds a separate 2.5 V wordline supply (VPP). Internally the array runs at roughly one eighth of the data rate seen at the pins, moving two words per clock edge on each DQ pin.

Family positioning, in one line each:

Tech	Prefetch	Distinctives
DDR2 SDRAM	4n	1.8 V, ODT, posted CAS, 4-bit prefetch
DDR3 SDRAM	8n	1.5 V, fly-by CA, ZQ calibration, write leveling
DDR4 SDRAM	8n	1.2 V + VPP 2.5 V, bank groups, higher data rates
DDR5 SDRAM	16n	1.1 V, two 32-bit channels per DIMM, on-die ECC

2.2 Key features of DDR4

- Bank groups: x4/x8 devices have four groups of four banks each (16 banks total); x16 devices have two groups of four banks each (8 banks total). This is the headline change: timing splits into “short” (different group) and “long” (same group) versions - tCCD_S/tCCD_L, tRRD_S/tRRD_L, and tWTR_S/tWTR_L.
- 8n prefetch retained: every column access still moves eight words internally.
- Burst length 8 (BL8) or burst chop 4 (BC4). BC4 can be selected on-the-fly through A12/BC_n during a column command, or fixed in the mode register.
- Gear-down mode: command/address timing can run in 1N (half-rate, default) or 2N (quarter-rate) mode, selected through MR3.
- Fine-granularity refresh: refresh can be configured as 1x, 2x, or 4x mode, with corresponding tRFC1/tRFC2/tRFC4 recovery times and tREFI intervals of 7.8 us, 3.9 us, and 1.95 us at normal temperature.
- Refresh is all-bank only; JESD79-4D does not define a per-bank refresh command (REFpb).
- Read and write data bus inversion (DBI) is supported on x8 and x16 parts, controlled through MR5.
- Write CRC: a CRC word is appended to write bursts; CRC errors are reported as a pulse on ALERT_n. Enabling CRC adds one clock to effective write latency.
- CA parity: an even parity input (PAR) covers the command/address bus; parity errors are flagged on ALERT_n and logged in MPR page 1. Parity latency (PL) is added to read and write latency.
- CAL mode: command/address latency delays chip-select assertion relative to the command to save power.
- Programmable read preamble (1 tCK or 2 tCK) and write preamble (1 tCK or 2 tCK), selected in MR4. Two-clock preambles are valid at DDR4-2400 and above.
- DQ Vref training: MR6 lets the host set the internal DQ reference voltage during training.

- Multi-purpose register (MPR) pages: four pages are available; page 0 is for training patterns, page 1 logs CA parity errors, page 2 exposes mode-register readback, and page 3 is vendor-specific.
- DLL-on is the normal mode; DLL-off mode is optional and limits speeds.
- Additive latency (AL) can be 0, CL-1, or CL-2.
- IO organizations are x4, x8, and x16; an x32 ballout is also described for specialized packages but is outside the standard monolithic density scope.

2.3 Densities and speed grades

Standard monolithic densities run from 2 Gb to 16 Gb in x4, x8, and x16 organizations. x4/x8 parts use 16 banks (four groups of four); x16 parts use 8 banks (two groups of four).

Speed grades span DDR4-1600 through DDR4-3200, including 1866, 2133, 2400, 2666, and 2933. Each grade carries a letter suffix that identifies the speed bin (for example, DDR4-1600J/K/L are different CL-tRCD-tRP bins).

2.4 The simplified drill model

Real parts have far more rows and columns than the model below. All examples in this book use a deliberately tiny geometry so bank state is obvious at a glance, and this is the first drill model that includes bank groups:

- Bank groups: BG0, BG1 (two groups)
- Banks per group: B0-B3 in BG0, B4-B7 in BG1
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

This model is used by the ddr_drills training app. Every command trace, walkthrough, and timing example in later chapters uses it. Real address maps are far larger; the scheduling rules are identical.

Source: JESD79-4D sections 1, 2.7, 2.8, 3.2, 4.9, 4.10, 4.11, 4.13, 4.15, 4.16, 4.17, 4.18, 4.20

3 Organization

3.1 Banks, rows, columns

A DDR4 device is organized as one or more bank groups, and each bank group contains four independent banks. Each bank is a 2D array addressed by row then column. An Activate command copies one row into that bank's sense amplifiers (the open row or page); reads and writes then hit the sense amplifiers at column granularity. A bank must be precharged before a different row in the same bank can be opened.

Addressing choices by density and IO width:

Density	Org	Bank groups	Banks per group	Row address	Column address	Page size
2 Gb	x4	4	4	A0-A14	A0-A9	512 B
2 Gb	x8	4	4	A0-A13	A0-A9	1 KB
2 Gb	x16	2	4	A0-A13	A0-A9	2 KB
4 Gb	x4	4	4	A0-A15	A0-A9	512 B
4 Gb	x8	4	4	A0-A14	A0-A9	1 KB
4 Gb	x16	2	4	A0-A14	A0-A9	2 KB
8 Gb	x4	4	4	A0-A16	A0-A9	512 B
8 Gb	x8	4	4	A0-A15	A0-A9	1 KB
8 Gb	x16	2	4	A0-A15	A0-A9	2 KB
16 Gb	x4	4	4	A0-A17	A0-A9	512 B
16 Gb	x8	4	4	A0-A16	A0-A9	1 KB
16 Gb	x16	2	4	A0-A16	A0-A9	2 KB

The two bank-address pins (BA0-BA1) select the bank inside a group; the bank- group pins (BG0-BG1 for x4/x8, BG0 for x16) select the group.

3.2 Pinout highlights versus DDR3

The command/address bus was reorganized and several new pins appeared:

Pin	Role
ACT_n	Activation command input; when low, RAS_n/A16, CAS_n/A15, WE_n/A14 carry row-address bits
BG0-BG1	Bank-group selection (x4/x8 have both; x16 has only BG0)
PAR	Even-parity input for the command/address bus
ALERT_n	Active-low alert output for CRC/parity errors; also a connectivity-test input
VPP	2.5 V activating (wordline) supply, in addition to 1.2 V VDD/VDDQ
TEN	Connectivity-test mode enable (required on x16, optional on x4/x8 8 Gb and above)

3.3 Bank group timing split

Because banks in different groups can operate more independently, several timing parameters have short and long forms:

Symbol	Applies between
tCCD_S	Consecutive column commands to different bank groups
tCCD_L	Consecutive column commands to the same bank group
tRRD_S	Consecutive Activate commands to different bank groups
tRRD_L	Consecutive Activate commands to different banks in the same group
tWTR_S	Write followed by read to a different bank group
tWTR_L	Write followed by read to the same bank group

tFAW still limits any rolling window of four Activates.

3.4 Addressing

The row address is supplied with the Activate command on A0-A17 (the exact width depends on density and organization). The column address is supplied with the Read or Write command. A10 is not a column bit; it is the auto- precharge flag (AP). A12/BC_n chooses between BL8 and BC4 on-the-fly during a column command when OTF is enabled; otherwise the burst mode is fixed in the mode register.

Page size formula: page size = 2^COLBITS x ORG / 8 bytes, per bank. Wider organizations use fewer column bits to keep the per-bank page at roughly 512 B (x4), 1 KB (x8), or 2 KB (x16).

3.5 Prefetch and IO width

The 8n prefetch means one column access pulls eight words per DQ pin group from the array in parallel. Consequences:

- The smallest useful burst is four data words per pin (BC4); BL8 uses the full eight-word fetch.
- The array core runs at one eighth of the pin data rate.
- A BC4 burst transfers the first four words of the prefetch and discards the second four.

IO organizations are x4, x8, and x16. x4 uses one DQS pair for four DQ pins; x8 uses one DQS pair for eight DQ pins; x16 has two DQS pairs, one for each byte lane. x8 parts also support an optional termination data strobe pair, TDQS/TDQS#, when enabled in MR1. A non-standard x32 ballout is also described for specialized packages.

3.6 Ranks and stacked packages

The spec covers single-die, dual-die, and quad-die packages. A stacked package presents multiple ranks behind one channel: each rank gets its own chip select (and its own ODT and CKE in multi-rank ballouts), shares the address/command and data busses, and is otherwise a complete independent DDR4 device. Rank-to-rank traffic adds bus-turnaround constraints beyond single-rank rules.

3.7 The drill model restated

This book's examples use two bank groups (BG0 and BG1), four banks per group (B0-B3 in BG0, B4-B7 in BG1), eight rows per bank (R0-R7), and eight columns per row (C0-C7). One bank's state at a glance:

Bank	State	Open row
B0	idle	-
B5	active	R1

Source: JESD79-4D sections 2.7, 2.8, 2.9, 3.2

4 Acronyms and Terms

Terms used with precise meanings throughout this book.

Term	Meaning
ACT	Bank Activate command: opens a row in a bank
AL	Additive latency: clocks a posted CAS is held inside the DRAM (MR1)
ALERT_n	Active-low alert pin: reports CRC and CA parity errors
AP	Auto precharge flag, carried on address bit A10 with RD/WR
BA	Bank address pins (BA0-BA1)
BC4	Burst chop 4: a four-word burst from the first half of the 8n prefetch
BG	Bank group address pins (BG0-BG1 for x4/x8, BG0 for x16)
BL	Burst length: 8 words (BL8) or 4 words via burst chop (BC4)
CAL	Command/address latency: delay from CS_n assertion to valid command (MR4)
CAS	Column Address Strobe (command pin); shorthand for a column command
CA parity	Even parity checked over the command/address bus (MR5)
CKE	Clock enable; gates power-down and self-refresh entry/exit
CL	CAS latency: read-command-to-data delay after the posted hold (MR0)
CRC	Cyclic redundancy check appended to write bursts (MR2)
CWL	CAS write latency: write-command-to-

Term	Meaning
	first-DQS delay (MR2)
DBI	Data bus inversion: inverts a byte lane when it saves power (MR5)
DES	Deselect command (CS# high)
DLL	Delay-locked loop aligning read DQS to CK
DM	Data mask pin (write data masking, per byte)
DQS	Data strobe; source-synchronous clock for DQ in both directions
FGR	Fine-granularity refresh: 1x/2x/4x refresh modes (MR3)
Gear-down	1N/2N command/address rate mode (MR3)
MPR	Multi-purpose register: four pages used for training and logging
MR / MRn	Mode Register n (0-6)
MRS	Mode Register Set command
NOP	No Operation command
ODT	On-die termination (Rtt on DQ/DQS/DM, per MR1)
OTF	On-the-fly: A12/BC_n selects BL8 or BC4 during a column command
PAR	Command and address parity input pin
PRE	Precharge command (A10=0: one bank; A10=1: all banks)
RD / WR	Read / Write commands (A10=0)
RDA / WRA	Read / Write with auto precharge (A10=1)
REF	Auto refresh command (all-bank, with 1x/2x/4x rate options)

Term	Meaning
RESET_n	Active-low asynchronous reset pin
RL	Read latency = AL + CL (+ PL when CA parity is enabled)
Rtt_NOM	Nominal on-die termination strength
Rtt_PARK	Parking termination strength (MR5)
Rtt_WR	Dynamic ODT strength used during writes
tCCD_L	Column-to-column delay for commands to the same bank group
tCCD_S	Column-to-column delay for commands to different bank groups
tRRD_L	Activate-to-Activate delay for banks in the same bank group
tRRD_S	Activate-to-Activate delay for Activates to different bank groups
tWTR_L	Write-to-read delay for reads in the same bank group
tWTR_S	Write-to-read delay for reads in a different bank group
VPP	2.5 V DRAM activating (wordline) supply
VrefDQ	Internal DQ reference voltage, trained via MR6
WL	Write latency = AL + CWL (+ PL when CA parity is enabled)
ZQ	Reference pin for ZQ calibration
ZQCL / ZQCS	ZQ calibration long / short commands
Bank group	A collection of banks that share the “long” timing path
Page / open row	The row currently held in a bank’s sense amplifiers

Command trace notation used in Chapters 3 and 4:

```
ACT B3 R5      # open row R5 in bank B3
RD  B3 C2      # read burst starting at column C2
WR  B1 C0      # write burst starting at column C0
RDA B2 C4      # read with auto precharge
PRE B3         # precharge bank B3
PREA           # precharge all banks
REF            # auto refresh
ZQCS / ZQCL    # ZQ calibration short / long
--- (tWTR bubble) ---   # forced idle bus time, reason named
```

Source: JESD79-4D sections 2.7, 2.8, 3.1, 3.2, 4.9, 4.10, 4.11, 4.13, 4.15, 4.16, 4.17, 4.18, 4.20

5 Power-Up and Initialization

DDR4 must be brought up through a fixed initialization flow. The mode registers have no reliable power-up defaults, so every defined register must be written before normal traffic is issued. The flow below covers cold power-up and a reset while power remains stable.

5.1 Cold start sequence

1. Apply power with RESET# below 0.2 x VDD and the TEN input also below 0.2 x VDD. Other input levels may be undefined during this phase. RESET# must stay low for at least 200 us once the supplies are stable; TEN must stay low for at least 700 us. CKE must be driven low at least 10 ns before RESET# rises.
2. Keep the voltage ramp within the required windows: VDD and VDDQ are nominally 1.2 V, VPP is nominally 2.5 V; VPP must ramp at the same time or earlier than VDD and must be equal to or higher than VDD at all times; VDD must be greater than or equal to VDDQ and their difference must stay below 0.3 V; the ramp from 300 mV to VDDmin must take no more than 200 ms.
3. Release RESET#, then wait 500 us before raising CKE. During this window the DRAM performs internal state initialization without needing a clock.
4. Start CK/CK# and let it stabilize for at least 10 ns or 5 tCK, whichever is longer, before CKE goes high. A Deselect command must be registered before the CKE rising edge. CKE must stay high without interruption for the rest of initialization, through tDLLK and tZQinit.
5. Keep ODT high-impedance from power-up until CKE is high. If RTT_NOM will be enabled in MR1, hold the ODT pin low from tIS before CKE rises until the end of initialization; otherwise it may be tied low or high but must not toggle.
6. After CKE is registered high, wait tXPR = max(tXS, 5 x tCK) before the first MRS command.
7. Issue MRS commands in this order: MR3, MR6, MR5, MR4, MR2, MR1, MR0. MR0 must include the DLL reset bit (A8 = 1).
8. Issue ZQCL to begin ZQ calibration.
9. Wait until both tDLLK and tZQinit have expired; the device is then ready for read/write training, which includes VrefDQ training and write leveling.

The gaps between commands look like this:

```
... CKE high ... MRS(MR3) ... MRS(MR6) ... MRS(MR5) ... MRS(MR4) ...
MRS(MR2) ... MRS(MR1) ... MRS(MR0) ... ZQCL ... ready
         |tXPR| tMRD |  tMRD  |  tMRD  |  tMRD  |  tMRD  |  tMRD  |
tMOD  |     tZQinit     |

Deselect commands must fill every gap from the first MRS through ZQCL.
```

Key timing values:

Symbol	Value	Meaning
tMRD	8 nCK	Minimum gap between consecutive MRS commands
tMOD	max(24 nCK, 15 ns)	Minimum gap from an MRS command to a non-MRS command
tDLLK	597 / 768 / 1024 nCK	DLL lock time; value scales with data rate and is programmed in MR6
tZQinit	spec-defined	Long ZQ calibration interval after reset

5.2 Reset with stable power

If power is already stable and only a reset is required, the sequence is shorter:

1. Drive RESET# below 0.2 x VDD for at least tPW_RESET (1 us). CKE must be low at least 10 ns before RESET# rises.
2. Continue from cold-start step 3: the 500 us wait, clock stabilization, CKE high, and the MRS/ZQCL sequence.
3. After tDLLK and tZQinit expire, the device is ready for training.

5.3 Optional features during initialization

- Gear-down mode (MR3 A3): the DRAM powers up in 1N (1/2 rate) mode. To use 2N (1/4 rate) mode, issue a low-frequency MRS to set MR3 A3 = 1, send a 1N sync pulse, then wait tSYNC_GEAR and tCMD_GEAR before continuing the normal initialization sequence in 2N mode. CAL and CA parity must be disabled before the gear-down MRS and may be re-enabled afterward.
- CA parity mode (MR5 A2:A0): when enabled, MRS-to-MRS and MRS-to-command delays become tMRD_PAR = tMOD + PL and tMOD_PAR = tMOD + PL, where PL is the programmed parity latency.
- CAL mode (MR4 A8:A6): when enabled, MRS-to-command delays become tMOD_CAL = tMOD + tCAL and MRS-to-MRS delays become tMRD_CAL = tMOD + tCAL.

5.4 Notes that bite

- The seven mode registers are not initialized to defined values by hardware; they must all be written explicitly. Skipping one leaves undefined settings.
- Every MRS command requires all banks precharged and idle, tRP satisfied, all bursts finished, and CKE high.
- The DLL must be enabled (MR1 A0 = 1) before it is reset (MR0 A8 = 1). After any DLL reset, wait tDLLK before issuing a Read or any synchronous ODT operation.
- ZQ calibration initializes the output driver and ODT impedance; it is required before normal operation, not optional.
- During initialization, ODT is not active even if the eventual MR1 setting will enable RTT_NOM. Do not rely on termination until after tDLLK and tZQinit.
- Some MRS changes are exempt from plain tMRD/tMOD and have their own settling rules; these include gear-down entry, CA parity, CAL, per-DRAM addressability, VrefDQ training, and maximum power saving mode.

Source: JESD79-4D sections 3.3, 3.4.1, 4.15, 4.17, 4.18

6 Mode Registers

DDR4 provides seven mode registers selected by the BG0, BA1, and BA0 pins during an MRS command. BG1 and A17 are reserved and must be driven to 0. Register values are undefined after power-up, so all seven must be written during initialization. Reprogramming later is allowed whenever all banks are idle.

Register select encoding:

BG0	BA1	BA0	Register
0	0	0	MR0
0	0	1	MR1
0	1	0	MR2
0	1	1	MR3
1	0	0	MR4
1	0	1	MR5
1	1	0	MR6
1	1	1	RCW1 (ignored by DRAM)

6.1 MR0 (BG0=0, BA1=0, BA0=0)

Field	Bits	Encoding
Burst length	A1:A0	00 = fixed BL8, 01 = on-the-fly BC4/BL8 via A12, 10 = fixed BC4, 11 = reserved
Read burst type	A3	0 = sequential, 1 = interleaved
CAS latency	A12, A6:A4, A2	see CL table
Test mode	A7	0 = normal, 1 = manufacturer test only
DLL reset	A8	1 = reset (self-clearing)
WR / RTP for auto-precharge	A13, A11:A9	A[11:9] selects 10/12/14/16/18/20/24/22 cycles; A13=1 adds 26 cycles

CL encoding:

A12	A6:A4	A2	CL
0	000	0	9
0	000	1	10
0	001	0	11
0	001	1	12
0	010	0	13
0	010	1	14
0	011	0	15
0	011	1	16
0	100	0	18
0	100	1	20
0	101	0	22
0	101	1	24
0	110	0	23

A12	A6:A4	A2	CL
0	110	1	17
0	111	0	19
0	111	1	21
1	000	0	25
1	000	1	26

Other combinations are reserved or device-specific. Overall read latency is RL = AL + CL. Program WR to a value equal to or greater than tWRmin in clock cycles; WR is used with tRP to determine tDAL for auto-precharge writes.

6.2 MR1 (BG0=0, BA1=0, BA0=1)

Field	Bits	Encoding
DLL enable	A0	0 = disabled, 1 = enabled (opposite sense from DDR3)
Output driver impedance	A2:A1	00 = RZQ/7, 01 = RZQ/5, others reserved
RTT_NOM	A10:A8	000 = off, 001 = RZQ/4, 010 = RZQ/2, 011 = RZQ/6, 100 = RZQ/1, 101 = RZQ/5, 110 = RZQ/3, 111 = RZQ/7
Additive latency	A4:A3	00 = 0, 01 = CL-1, 10 = CL-2, 11 = reserved
Write leveling enable	A7	0 = off, 1 = enabled
TDQS enable	A11	0 = off, 1 = enabled (x8 only; disables DM)
Output disable (Qoff)	A12	0 = outputs enabled, 1 = DQ/DQS/DQS# disabled

When the DLL is disabled, synchronous ODT and dynamic ODT are not supported; keep ODT low and set Rtt_WR to off.

6.3 MR2 (BG0=0, BA1=1, BA0=0)

Field	Bits	Encoding
CAS write latency	A5:A3	000 = 9, 001 = 10, 010 = 11, 011 = 12, 100 = 14, 101 = 16, 110 = 18, 111 = 20; final value also depends on read/write preamble
Low-power auto self-refresh	A7:A6	00 = manual normal temp, 01 = manual reduced temp, 10 = manual extended temp, 11 = ASR mode
Dynamic ODT (Rtt_WR)	A11:A9	000 = off, 001 = RZQ/2, 010 = RZQ/1, 011 = Hi-Z, 100 = RZQ/3, others reserved
Write CRC enable	A12	0 = disabled, 1 = enabled

Overall write latency is WL = AL + CWL + PL, where PL is any programmed CA parity latency.

6.4 MR3 (BG0=0, BA1=1, BA0=1)

Field	Bits	Encoding
MPR operation	A2	0 = normal DRAM array, 1 = reads/writes redirected to MPR
MPR page selection	A1:A0	00 = page 0, 01 = page 1, 10 = page 2, 11 = page 3
Gear-down mode	A3	0 = 1N (1/2 rate), 1 = 2N (1/4 rate)
Per-DRAM addressability	A4	0 = disabled, 1 = enabled
Temperature sensor readout	A5	0 = disabled, 1 = enabled
Fine-granularity refresh	A8:A6	000 = fixed 1x, 001 = fixed 2x, 010 = fixed 4x, 101 = on-the-fly 2x, 110 = on-the-fly 4x, others reserved
Write CMD latency with CRC+DM	A10:A9	00 = 4 nCK, 01 = 5 nCK, 10 = 6 nCK, 11 = reserved
MPR read format	A12:A11	00 = serial, 01 = parallel, 10 = staggered, 11 = reserved

With MPR enabled, only RD/RDA and MPR write commands are legal until MPR is disabled. Page 0 holds a training pattern, page 1 logs CA parity errors, page 2 reflects mode-register readouts and temperature status, and page 3 is vendor-specific.

6.5 MR4 (BG0=1, BA1=0, BA0=0)

Field	Bits	Encoding
MBIST PPR	A0	0 = disabled, 1 = enabled
Maximum power saving mode	A1	0 = disabled, 1 = enabled
Temperature-controlled refresh range	A2	0 = normal, 1 = extended
Temperature-controlled refresh mode	A3	0 = disabled, 1 = enabled
Internal Vref monitor	A4	0 = disabled, 1 = enabled
Soft PPR	A5	0 = disabled, 1 = enabled
CS-to-CA latency (CAL)	A8:A6	000 = disable, 001 = 3, 010 = 4, 011 = 5, 100 = 6, 101 = 8, others reserved
Self-refresh abort	A9	0 = disabled, 1 = enabled
Read preamble training mode	A10	0 = disabled, 1 = enabled
Read preamble	A11	0 = 1 tCK, 1 = 2 tCK
Write preamble	A12	0 = 1 tCK, 1 = 2 tCK
Hard PPR	A13	0 = disabled, 1 = enabled

6.6 MR5 (BG0=1, BA1=0, BA0=1)

Field	Bits	Encoding
CA parity latency mode	A2:A0	000 = disabled, 001 = PL=4, 010 = PL=5, 011 = PL=6, others reserved
CRC error clear	A3	0 = clear, 1 = error
CA parity error status	A4	0 = clear, 1 = error
ODT input buffer in power-down	A5	0 = activated, 1 = deactivated
RTT_PARK	A8:A6	same encoding as RTT_NOM
CA parity persistent error	A9	0 = disabled, 1 = enabled
Data mask (DM)	A10	0 = disabled, 1 = enabled
Write DBI	A11	0 = disabled, 1 = enabled
Read DBI	A12	0 = disabled, 1 = enabled

When DM is enabled together with write CRC, an extra write-command latency is added as defined in MR3 A10:A9.

6.7 MR6 (BG0=1, BA1=1, BA0=0)

Field	Bits	Encoding
VrefDQ training value	A5:A0	valid training codes; step size is about 0.65 % per code
VrefDQ training range	A6	0 = range 1, 1 = range 2
VrefDQ training enable	A7	0 = normal operation, 1 = training mode
tCCD_L and tDLLK	A12:A10	000 = 4/597, 001 = 5/597, 010 = 6/768, 011 = 7/768, 100 = 8/1024 nCK; 101-111 reserved

The exact tCCD_L and tDLLK values depend on the operating data rate and the AC parameter tables.

6.8 Programming rules

- Precondition: all banks precharged and idle, tRP satisfied, all bursts finished, CKE high.
- An MRS command rewrites the whole addressed register; always drive every field intentionally, including unchanged ones.
- Wait tMRD (8 nCK) between consecutive MRS commands and tMOD (max(24 nCK, 15 ns)) before any non-MRS command (NOP/Deselect excluded).
- If CA parity or CAL is enabled, use tMRD_PAR / tMOD_PAR or tMRD_CAL / tMOD_CAL instead of the plain values.
- If RTT_NOM is enabled before or after the MRS, keep ODT low through the command and do not raise it until tMOD expires. If RTT_NOM is disabled, ODT is a do not care during MRS.
- Drive BG1, A17, and all reserved address bits to 0.
- MRS and DLL reset do not disturb stored array data.

Source: JESD79-4D sections 3.4.1, 3.5, 4.7, 4.15, 4.17, 4.18

7 Command Truth Tables

A command is latched on each rising CK edge from CS_n, ACT_n, RAS_n/A16, CAS_n/A15, WE_n/A14, the bank-group and bank addresses, plus the row or column address fields. H = high, L = low, X = do not care, V = a valid logic level. With ACT_n low the command pins serve as address bits A16-A14; with ACT_n high they serve as RAS_n, CAS_n, and WE_n.

7.1 Command truth table

Function	CKE prev	CKE cur	CS_n	ACT_n	RAS/A16	CAS/A15	WE/A14	BG	BA	A12/BC_n	A10/AP	Notes
MRS	H	H	L	H	L	L	L	BG	BA	OP code	OP code	BA/BG select MR0-MR6
REF (all-bank)	H	H	L	H	L	L	H	V	V	V	V	all banks idle
SRE (self-refresh entry)	H	L	L	H	L	L	H	V	V	V	V	REF encoding + falling CKE
SRX (self-refresh exit)	L	H	H	X	X	X	X	X	X	X	X	DES only (NOP allowed for gear-down/MPSM exit

Function	CKE prev	CKE cur	CS_n	ACT_n	RAS/A16	CAS/A15	WE/A14	BG	BA	A12/BC_n	A10/AP	Notes
												)
PRE single bank	H	H	L	H	L	H	L	BG	BA	V	L	closes bank selected by BG/BA
PREA all banks	H	H	L	H	L	H	L	V	V	V	H	closes every bank
ACT	H	H	L	L	RA	RA	RA	BG	BA	V	V	row on A[16:0] when ACT_n = L
WR fixed BL8 or BC4	H	H	L	H	H	L	L	BG	BA	V	L	A12 selects burst length when OTF

Function	CKE prev	CKE cur	CS_n	ACT_n	RAS/A16	CAS/A15	WE/A14	BG	BA	A12/BC_n	A10/AP	Notes
												is enabled
WRS4 (BC4 OTF)	H	H	L	H	H	L	L	BG	BA	L	L	BC4 on-the-fly
WRS8 (BL8 OTF)	H	H	L	H	H	L	L	BG	BA	H	L	BL8 on-the-fly
WRA	H	H	L	H	H	L	L	BG	BA	V	H	write + auto-precharge
RD fixed BL8 or BC4	H	H	L	H	H	L	H	BG	BA	V	L	A12 selects burst length when OTF is enabled

Function	CKE prev	CKE cur	CS_n	ACT_n	RAS/A16	CAS/A15	WE/A14	BG	BA	A12/BC_n	A10/AP	Notes
RDS4 (BC4 OTF)	H	H	L	H	H	L	H	BG	BA	L	L	BC4 on-the-fly
RDS8 (BL8 OTF)	H	H	L	H	H	L	H	BG	BA	H	L	BL8 on-the-fly
RDA	H	H	L	H	H	L	H	BG	BA	V	H	read + auto-precharge
NOP	H	H	L	H	H	H	H	V	V	V	V	bus filler, CS_n low
DES	H	H	H	X	X	X	X	X	X	X	X	bus filler, CS_n high
PDE (power	H	L	H	X	X	X	X	X	X	X	X	DES only

Function	CKE prev	CKE cur	CS_n	ACT_n	RAS/A16	CAS/A15	WE/A14	BG	BA	A12/BC_n	A10/AP	Notes
-down entry)												
PDX (power-down exit)	L	H	H	X	X	X	X	X	X	X	X	DES only
ZQCL	H	H	L	H	H	H	L	V	V	V	H	long ZQ calibration
ZQCS	H	H	L	H	H	H	L	V	V	V	L	short ZQ calibration
RESET	X	X	X	X	X	X	X	X	X	X	X	RESET# = L, asynchronous override

Key mnemonics:

- ACT = ACT_n low; the shared RAS/A16, CAS/A15, WE/A14 pins become address bits for the row.
- PRE/PREA = ACT_n and RAS/A16 low, CAS/A15 high, WE/A14 low; A10/AP chooses one bank vs all banks.
- RD = CAS/A15 low, WE/A14 high; WR = CAS/A15 and WE/A14 low; A10/AP adds auto-precharge.
- MRS = ACT_n, RAS/A16, CAS/A15, WE/A14 all low; BG1/BG0 and BA1/BA0 select the target mode register.
- REF = ACT_n, RAS/A16, CAS/A15 low and WE/A14 high; the same encoding with falling CKE becomes self-refresh entry.
- A12/BC_n is the burst-length control: H = BL8, L = BC4. It is ignored when the mode register fixes the burst length.

7.2 CKE truth table

Current state	CKE prev	CKE cur	Command at N	Result
Power-down	L	L	X	remain in power-down
Power-down	L	H	DES	exit power-down
Self-refresh	L	L	X	remain in self-refresh
Self-refresh	L	H	DES (NOP for gear-down/MPSM exit)	exit self-refresh
Bank(s) active	H	L	DES	active power-down entry
Reading / writing / precharging / refreshing	H	L	DES	power-down entry after the operation finishes
All banks idle	H	L	DES	precharge power-down entry
All banks idle	H	L	REF	self-refresh entry

7.3 Burst length, type and order

Burst length and on-the-fly selection are controlled by MR0 A[1:0] and A12/BC_n during read/write commands. MR0 A3 chooses sequential (0) or interleaved (1) burst type. Burst chop 4 returns only the first four UI; the remaining positions are don’t care on reads and ignored on writes. With write CRC enabled, fixed BL8 burst ordering is restricted to starting column A[2:0] = 000.

7.4 Data mask, DBI and TDQS

DDR4 uses one DM_n/DBI_n/TDQS_t pin per byte lane on x8 and x16 devices; x4 devices do not support DM or DBI. MR1 A11 enables TDQS on x8; when TDQS is on, DM and DBI are unavailable. MR5 A10 enables DM, A11 enables write DBI, and A12 enables read DBI. Write DBI and DM cannot both be enabled. On reads the DRAM inverts a byte and drives DBI_n low when more than half of the eight data bits are 0.

7.5 Not in the table

- No Burst Terminate command exists in DDR4 (DDR had one; DDR2 and later dropped it).
- No Off-Chip Driver Calibration (OCD) command exists in DDR4; OCD was removed.
- No per-bank refresh exists in JESD79-4D; a REF cycle always refreshes all banks.
- DBI, DM and TDQS are programmable mode-register features, not separate commands.

Source: JESD79-4D sections 4.1, 4.2, 4.3, 4.11, 4.16, 4.26

8 Commands: Rules and Constraints

What each command does and what must come next. Timing symbols are defined in Chapter 6; examples use the drill model (BG0 = B0-B3, BG1 = B4-B7, R0-R7, C0-C7).

8.1 ACT - Bank Activate

Opens row R in bank B so reads and writes can follow.

- ACT -> RD/WR same bank: wait tRCD, unless additive latency posts the command. With AL > 0 the RD/WR can be issued immediately, but internal CAS must not occur before tRCD.
- ACT -> ACT different bank group: tRRD_S.
- ACT -> ACT same bank group, different bank: tRRD_L.
- ACT -> ACT same bank: tRC (= tRAS + tRP), and the prior row must already be precharged.
- ACT -> PRE same bank: tRAS (the open row must remain active long enough).
- At most 4 ACTs in any rolling tFAW window. tFAW is independent of bank group.

8.2 RD / RDA - Read

Burst read starting at the given column. Data appears RL = AL + CL + PL clocks after the command, where PL is the CA parity latency if enabled. DQS begins toggling one or two cycles before the first data, selected by MR4 A[12:11] (1tCK or 2tCK read preamble).

- RD -> RD (different bank group): tCCD_S = 4 clocks.
- RD -> RD (same bank group): tCCD_L, programmed in MR0 A[12:10] (5-8 clocks depending on speed grade).
- RD -> PRE same bank: earliest spacing is AL + tRTP, also bounded by tRAS. tRTP = max(4 nCK, 7.5 ns).
- RDA (A10/AP = 1): auto-precharge starts internally after AL + tRTP, delayed if tRAS has not been met. Next ACT to that bank needs tRP from when precharge starts and tRC from the old ACT.
- RD -> WR (book symbol tRTW): JESD79-4D does not name tRTW; the figures give the command spacing as RL + BL/2 + 2 - WL clocks. For BL8 this is RL + tCCD_S + 2 - WL; for BC4 it is RL + tCCD_S/2 + 2 - WL.

8.3 WR / WRA - Write

Burst write. The first data is captured WL = AL + CWL + PL clocks after the command. DQS must satisfy tDQSS relative to CK.

- WR -> WR (different bank group): tCCD_S = 4 clocks.
- WR -> WR (same bank group): tCCD_L.
- WR -> RD: wait until the last write data has been absorbed and tWTR has passed. tWTR_S applies to different bank groups; tWTR_L applies to the same bank group. The reference point is the first rising CK after the last write data.
- WR -> PRE same bank: WL + BL/2 + tWR. tWR is the write recovery time (15 ns minimum, programmed in MR0 A[13,11:9]); violating it corrupts data.
- WRA (A10/AP = 1): internal auto-precharge starts after write recovery. The bank can be re-activated once (WRA + tRP) and tRC are met.
- Write CRC (MR5 A12 = 1): adds one UI to the write burst and extends WL by 1. When both CRC and DM are enabled, use tWR_CRC_DM and tWTR_S_CRC_DM / tWTR_L_CRC_DM in place of the normal values.
- Write DBI and DM cannot both be enabled.

8.4 PRE / PREA - Precharge

Closes the open row and restores data to the array. A10/AP = 0 closes the bank selected by BG/BA; A10/AP = 1 closes all banks.

- PRE -> ACT same bank: tRP.
- Illegal if it would truncate a burst still in flight.

8.5 REF - Refresh

One internal refresh step across all banks; the address counter is internal.

- Precondition: all banks precharged and tRP met.
- REF -> ACT or REF: tRFC (density-dependent).
- Fine granularity refresh (FGR) modes: 1x (tRFC1, tREFI1 = 7.8 us), 2x (tRFC2, tREFI2 = 3.9 us), and 4x (tRFC4, tREFI4 = 1.95 us) at normal temperature. Program MR3 A[8:6] for fixed or on-the-fly selection; REF encoding with BG0 selects the rate in on-the-fly mode.
- Scheduling slack for 1x mode: up to 8 REFs may be postponed (max gap 9 x tREFI), and up to 8 may be pulled in early. Scaling applies to 2x and 4x modes.
- DDR4 does not define per-bank refresh; every REF refreshes all banks.

8.6 Self-refresh entry / exit

Entry: REF encoding with CKE falling, all banks idle, ODT off. The DRAM keeps itself refreshed without needing CK; only CKE matters.

- Minimum stay: tCKESR, plus an internal refresh occurs within tCKE of entry.
- Exit: CK must be stable before CKE rises. DES during the exit window.
- tXS after exit before any non-read command; tXSDLL before any command that needs a locked DLL (RD/RDA). tXS = tRFC(min) + 10 ns.
- MR4 A9 selects self-refresh abort mode: when enabled, use tXS_Abort (tRFC4(min) + 10 ns) and the DRAM aborts any ongoing refresh without incrementing the counter.
- Before re-entering self-refresh, issue at least one REF.
- LP ASR (MR2 A[7:6]) selects the self-refresh temperature range.

8.7 Power-down

CKE low with DES. Two flavors: precharge PD (all banks closed) and active PD (a row open). The DLL is kept on in both cases.

- Exit to any valid command: tXP. tXP = max(4 nCK, 6 ns) for DLL-on exit.
- CKE low pulse must satisfy tCKE. The CKE transition windows need DES coverage (tCPDED).
- PD performs no refresh; stay bounded by refresh requirements, at most 9 x tREFI.
- Entry is not allowed during MRS, MPR, ZQCAL, DLL lock, or read/write; it is allowed during activation, precharge/auto-precharge, and refresh once those operations complete.

8.8 NOP and DES

Bus fillers. NOP keeps CS_n low; DES raises CS_n. Neither changes state. Use them to occupy forced gaps (tRRD, tCCD, turnaround bubbles, tCPDED windows).

8.9 RESET

RESET# low is an asynchronous override. It resets the device, aborts any operation, and forces outputs to High-Z. After RESET# rises the normal power-up or stable-power reset sequence must be followed (200 us at power-up or 100 us with stable power, then 500 us before CKE high, etc.). ODT is High-Z while RESET# is asserted and until CKE is registered high. tXPR = max(5 nCK, tRFC(min) + 10 ns).

8.10 MRS / mode-register updates

MRS writes MR0-MR6. All address fields of the target register are rewritten, even if only a subset changes.

- MRS -> MRS: tMRD (8 nCK).
- MRS -> any non-NOP/DES command: tMOD = max(24 nCK, 15 ns).
- MRS must be issued when all banks are idle, all bursts complete, and CKE high. If RTT_NOM is enabled, ODT must be held low before and during the MRS command until tMOD expires.
- DLL reset (MR0 A8 = 1) needs tMOD and then tDLLK before DLL-dependent commands.
- With CAL enabled, use tMOD_CAL = tMOD + tCAL and tMRD_CAL = tMOD + tCAL.

8.11 DLL-off mode and DLL on/off switching

DLL-off is entered by clearing MR1 A0. Maximum clock frequency is tCKDLL_OFF; CL=10 and CWL=9 are the required settings. CA parity is not allowed with DLL off. Read data timing uses RL = AL + (CL - 1) and a larger tDQSCK window. Switching DLL state requires entering self-refresh, changing frequency, and meeting tCKSRE/tCKSRX/tXS before normal commands resume.

8.12 Input clock frequency change

The clock may change only during self-refresh. Hold ODT and CKE low, wait tCKSRE after CKE falls, change frequency, provide a stable clock for tCKSRX, then exit. After exit and tXS, update CL/CWL/WR/gear-down/preamble/CAL/parity settings as needed. CA parity latency must be disabled before the frequency change and re-enabled afterward.

8.13 Write leveling entry / exit

Used to align DQS-DQS# to CK-CK# across fly-by routing.

- Entry: MRS to MR1 with A7 = 1; output buffers of other ranks disabled via MR1 A12 = 1. Only DES and another MRS are valid in leveling mode.
- The DRAM asynchronously feeds back the sampled CK level on DQ. The controller sweeps DQS delay until it sees a 0 -> 1 transition.
- Exit: stop driving DQS, drive ODT low, then MRS MR1 with A7 = 0. Wait tMOD before normal commands; MRS-to-MRS spacing is tMRD.

8.14 MPR read flow

The Multi-Purpose Register outputs a predefined calibration pattern or accepts training writes.

- Precharge all banks and wait tRP.
- MRS to MR3 with A2 = 1 (and A[1:0] selecting the page) to enter MPR mode.
- Wait tMRD and tMOD.
- Issue RD/RDA with A[1:0] = 00 and A2 = 0 for BL8; only RD/RDA/WR/WRA/DES/REF/MRS/RESET are legal in MPR mode. RDA/WRA behave like plain reads/writes (auto-precharge ignored). Only 1x refresh is allowed.
- After the last MPR burst, wait tMPRR, then MRS to MR3 with A2 = 0 to exit.
- Wait tMRD and tMOD before normal array commands.

8.15 Gear-down mode

Gear-down (2N command timing) is selected by MR3 A3. It can only be entered during initialization or after self-refresh exit. The sequence is: a low-frequency 1N MRS to set MR3 A3 = 1, then a 1N sync pulse (NOP) lasting an even number of clocks, then normal 2N operation after tCMD_GEAR. CL, CWL, WR/RTP, AL, CAL and PL must all be programmed to even values. CAL and CA parity must be disabled before the gear-down MRS and can be re-enabled after tSYNC_GEAR and tCMD_GEAR. Self-refresh resets the DRAM back to 1N mode.

8.16 CAL mode

Command Address Latency (CAL) is a power-saving feature controlled by MR4 A[8:6]. CS_n is asserted tCAL cycles before the command/address is valid, giving the DRAM time to enable its receivers. Valid settings are 3, 4, 5, 6, and 8 clocks. With CAL enabled, MRS-to-command spacing becomes tMOD_CAL = tMOD + tCAL and MRS-to-MRS spacing becomes tMRD_CAL = tMOD + tCAL. In geardown mode, odd CAL values are rounded up to the next even integer.

8.17 CRC error handling

Write CRC is enabled by MR5 A12. The DRAM compares the received checksum against its own calculation. On mismatch it drives ALERT# low for at least 6 clocks (CRC ALERT_PW) within tCRC_ALERT (max 13 ns). It also sets MR5 A3 (CRC Error Clear) and a status bit in MPR page 1. The controller must retry the write and maintain coherence. CRC and DM together block the write to the array on error; with CRC alone the DRAM may write bad data, so the controller retries.

8.18 CA parity error handling

Turn on CA parity by writing a non-zero parity latency (PL) into MR5 A[2:0]. The PAR bit covers ACT_n, RAS/A16, CAS/A15, WE/A14, BG, BA, and A[17:0]; unused address pins are treated as 0 inside the DRAM. The parity convention is even. After detecting a parity error the DRAM drops the bad command, drives ALERT# low for at least tPAR_ALERT_PW, records the errant frame in MPR page 1, sets MR5 A4 (Parity Error Status), precharges all banks once tRAS_min is satisfied, and accepts normal commands again only after the status bit is cleared. With persistent parity error mode enabled (MR5 A9 = 1), parity checking resumes automatically once the alert pulse ends.

Source: JESD79-4D sections 4.1, 4.2, 4.4, 4.6, 4.7, 4.9, 4.10, 4.11, 4.12, 4.15, 4.16, 4.17, 4.18, 4.19, 4.22, 4.23, 4.24, 4.25, 4.26, 4.27, 4.28

9 Walkthrough 1: AXI4 Read, Row Hit

9.1 Controller configuration used in all Chapter 4 walkthroughs

Drill model geometry (B0-B7, R0-R7, C0-C7), BL8, additive latency off (AL=0; AL is a programmable field in MR1, but this drill keeps it disabled). Two bank groups are used: BG0 = {B0,B1,B2,B3}, BG1 = {B4,B5,B6,B7}.

Parameter	Value (clocks)
CL / CWL / AL / RL / WL	10 / 10 / 0 / 10 / 10
tRCD / tRP / tRAS / tRC	5 / 5 / 15 / 20
tRRD_S / tRRD_L / tFAW	2 / 3 / 12
tCCD_S / tCCD_L	4 / 6
tWTR_S / tWTR_L	2 / 4
tWR / tRTP	12 / 6

The AXI4 data bus is assumed as wide as one DRAM burst, so one AXI4 beat maps to one DRAM burst word: an AXI4 burst of 8 beats is exactly one BL8 DRAM burst. Real controllers slice and dice; the drill model does not. Burst chop (BC4) versus BL8 can be chosen on-the-fly via A12 when MR0 allows it; these walkthroughs keep A12 high for BL8.

9.2 Scenario

Initial bank state: B3 has row R5 open. All other banks idle.

AXI4 master issues: ARADDR mapping to bank B3, row R5, column C2, ARLEN=7 (8 beats), ARSIZE matching full bus width.

9.3 Command trace

```
RD  B3 C2 (BG0)  # row R5 already open in B3 - no ACT needed, go
straight to CAS
```

One command. That is the whole point of a row hit.

Timeline:

```
clk:      0        1    2    3    4    5    6    7    8    9   10   11
12   13   14   15   16
CMD:      RD
DQS:                        pre  ----DQS toggling----
DQ:                              D0   D1   D2   D3   D4   D5   D6   D7
```

Data appears RL = AL + CL = 10 clocks after the RD command. AL is programmed to zero in this drill, so the CAS is executed immediately; there is nothing to post against. DDR4 computes write latency as AL + CWL, not the RL-1 shortcut used in DDR2.

9.4 What happens on the AXI side

- AR handshake completes; the controller's address mapper splits the address into {BG0, B3, R5, C2}.
- The bank machine for B3 reports R5 open -> issue RD immediately, skipping ACT entirely.
- At RL the DRAM drives DQS/DQ; the PHY returns 8 data words; the controller streams them out as 8 AXI R beats with RLAST on the final word.

9.5 Scheduling takeaways

- Row-hit reads are the cheapest DRAM transaction: 1 command, latency RL.
- The bank stays open afterwards (A10=0). A follow-up access to R5 is another hit; an access to a different row of B3 now costs a PRE + ACT (see Walkthrough 2).
- A controller doing FR-FCFS scheduling reorders the queue exactly to manufacture this case.

Source: JESD79-4D sections 2.8, 4.19, 4.22, 4.23, 4.24

10 Walkthrough 2: AXI4 Read, Row Miss

Same controller configuration as Walkthrough 1 (CL=10, CWL=10, AL=0, RL=10, WL=10, BL8; tRCD=5, tRP=5, tRAS=15). Bank groups are BG0 = {B0,B1,B2,B3} and BG1 = {B4,B5,B6,B7}.

10.1 Scenario

Initial bank state: B5 has row R6 open (from earlier traffic). AXI4 read arrives: ARADDR maps to bank B5, row R2, column C1, ARLEN=7.

Row R2 is not the open row -> row miss. The bank must be closed and the right row opened before the read.

10.2 Command trace

```
PRE B5 (BG1)     # close R6 - wrong row is open in B5, precharge
before re-activating
--- (tRP = 5) ---# sense amps need tRP to finish restoring before the
next ACT
ACT B5 R2 (BG1)  # open the requested row R2
--- (tRCD = 5) ---# row decode needs tRCD before a CAS command is
legal
RD  B5 C1 (BG1)  # AL=0: no posted CAS, issue exactly at ACT+tRCD
```

Timeline:

```
clk:      0     1    2    3    4     5     6     7     8     9    10
11   12   13   14   15   16   17   18   19   20   21   22   23   24
25
CMD:      PRE                               ACT                     RD
               |<----- tRP=5 ----->|        |<----- tRCD=5 ----->|
DQ:
D0   D1   D2   D3   D4   D5   D6   D7
```

- PRE at clk 0, ACT legal at clk 5 (tRP).
- RD issued at clk 10 (tRCD after ACT at clk 5).
- Read data starts at clk 10 + RL = clk 20.

10.3 Cost accounting

Row miss penalty = tRP + tRCD of pure dead time on this bank before data can flow: 10 clocks here, versus 0 for a hit. With AL=0, posted CAS is off, so the RD command slot is spent at ACT+tRCD; nothing hides the latency.

10.4 AXI side

- The AR channel accepts the request; the controller cannot produce R beats until the DRAM data comes back, so the master just sees a longer RVALID latency. Nothing on the AXI interface changes - all of this is hidden inside the controller.
- If the master had a second read outstanding to a different (idle) bank, a good scheduler would run that one first during the tRP/tRCD bubble.

10.5 Variation: read with auto precharge (RDA)

If the controller predicts no further use of R2 soon, it issues `RDA B5 C1 (BG1)` (A10=1) instead. The bank then precharges itself after the read burst completes and tRTP is satisfied, and is idle for the next tenant without a separate PRE command slot. In DDR4 the minimum read-to-precharge spacing for auto-precharge is AL + tRTP clocks after the RD command, delayed if tRAS is not yet met.

Source: JESD79-4D sections 2.8, 4.19, 4.22, 4.23, 4.24, 4.24.3

11 Walkthrough 3: AXI4 Write to an Idle Bank

Same controller configuration (CL=10, CWL=10, AL=0, RL=10, WL=10, BL8; tRCD=5, tRP=5, tWR=12). Bank groups are BG0 = {B0,B1,B2,B3} and BG1 = {B4,B5,B6,B7}.

11.1 Scenario

Initial state: all banks idle. AXI4 write arrives: AWADDR maps to bank B1, row R4, column C0, AWLEN=7, followed by 8 W-channel beats.

An idle bank needs no precharge - miss-to-idle is cheaper than miss-to-open-row by tRP.

11.2 Command trace

```
ACT B1 R4 (BG0)  # open the row; bank was idle so no PRE needed
--- (tRCD = 5) ---# wait for row decode before issuing CAS
WR  B1 C0 (BG0)  # AL=0; issued exactly at ACT+tRCD
--- (WL = 10) --- # write latency: first DQS edge lands 10 clks after
the WR command
--- (tWR = 12) ---# write recovery: last data must soak into the array
before PRE
PRE B1 (BG0)     # close the row (only if the controller wants the
bank back)
```

Timeline:

```
clk:      0     1     2    3    4    5     6    7    8    9   10   11
12   13   14   15   16   17   18   19   20   21   22   23   24   25
26   27   28   29   30   31
CMD:      ACT                           WR
PRE
                                   |<---- WL=10 ---->|
DQS:                                          pre -DQS toggling-
DQ:                                            D0   D1   D2   D3   D4
D5   D6   D7

|<--- tWR=12 --->|
PRE legal at clk 5 + 10(WL) + 4(BL/2) + 12(tWR) = clk 31 from ACT,
i.e. WL + BL/2 + tWR = 26 clocks after the WR command.
```

11.3 AXI side

- AW and W are independent channels; the controller buffers the 8 W beats (or streams them) while the DRAM side runs ACT + WR.
- BREADY goes back to the master once the controller has accepted responsibility for the data - typically after the write is safely in flight to the DRAM, not after tWR completes.
- Write data is strobed in with DQS driven by the controller/PHY; tDQSS alignment to CK is the PHY's job.

11.4 Write recovery is the trap

tWR is not a turnaround nicety; it is array physics. Precharging B1 before tWR expires destroys the data just written - the bits never finish committing from the sense amps to the cells. This is why WR is a mode-register field: the DRAM itself uses the programmed WR value to time auto precharge after a WRA, and tDAL = WR + roundup(tRP/tCK) is the real “write to next activate” cost when auto precharge is used.

11.5 Comparison against the read miss

	Read miss (walkthrough 2)	Write to idle bank
Commands before data	PRE, ACT, RD	ACT, WR
Dead time before data	tRP + tRCD	tRCD (hidden by WL buffering)
Tail cost	none (bank left open)	tWR before PRE

Source: JESD79-4D sections 2.8, 4.19, 4.22, 4.23, 4.25, 4.25.5

12 Walkthrough 4: Mixed Traffic and Bus Turnarounds

Same controller configuration (CL=10, CWL=10, AL=0, RL=10, WL=10, BL8; tWTR_S=2, tWTR_L=4, tWR=12, tCCD_S=4, tCCD_L=6, tRRD_S=2, tRRD_L=3, tFAW=12). Bank groups are BG0 = {B0,B1,B2,B3} and BG1 = {B4,B5,B6,B7}.

The shared DQ bus can only face one way at a time. Read-to-write and write-to-read direction changes cost idle bus clocks - the turnaround bubbles - and they are where bandwidth goes to die in mixed traffic. Bank groups add a second lever: same-group spacing is tCCD_L/tRRD_L/tWTR_L; different-group spacing is the shorter tCCD_S/tRRD_S/tWTR_S.

12.1 Scenario

Seven AXI4 requests, in arrival order:

1. Read B0 R5 C2 (B0 has R5 open - hit, BG0)
2. Read B1 R5 C6 (B1 has R5 open - hit, BG0, same group as #1)
3. Read B5 R5 C2 (B5 has R5 open - hit, BG1, different group from #2)
4. Write B5 R5 C0 (BG1, same group as #3)
5. Read B2 R5 C4 (B2 has R5 open - hit, BG0, different group from #4)
6. Write B2 R5 C7 (BG0, same group as #5)
7. Read B3 R5 C5 (B3 has R5 open - hit, BG0, same group as #6)

12.2 Command trace

```
RD  B0 C2 (BG0)  # hit: data flows immediately, RL=10
--- (RD B0 -> RD B1: tCCD_L = 6, same bank group) ---
RD  B1 C6 (BG0)  # same group as B0 -> long CAS-to-CAS spacing
--- (RD B1 -> RD B5: tCCD_S = 4, different bank group) ---
RD  B5 C2 (BG1)  # different group -> short spacing; groups interleave
--- (RD B5 -> WR B5: tRTW = 6, read burst must clear the bus) ---
WR  B5 C0 (BG1)  # write to same group as the read
--- (WR B5 -> RD B2: tWTR_S = 2, different bank group -> 16 clk total)
---
RD  B2 C4 (BG0)
--- (RD B2 -> WR B2: tRTW = 6) ---
WR  B2 C7 (BG0)
--- (WR B2 -> RD B3: tWTR_L = 4, same bank group -> 18 clk total) ---
RD  B3 C5 (BG0)
```

Timeline (clock numbers follow the spacing rules above):

```
clk:     0    1    2    3    4    5    6    7    8    9   10   11   12
13   14   15   16   17   18   19   20
CMD:     RD   .    .    .    .    .   RD   .    .    .   RD   .    .
.    .    .   WR   .    .    .    .
         |<--tCCD_L=6-->|      |<--tCCD_S=4-->|      |<--- tRTW = 6
--->|
bus:     ........ D0-D7 ............... D0-D7 ....... D0-D7 ....... p
D0-D7

clk:    21   22   23   24   25   26   27   28   29   30   31   32   33
34   35   36   37   38   39   40   41
CMD:     .    .    .    .    .    .    .    .    .    .    .   RD   .
.    .    .    .   WR   .    .    .
                              |<-------- WL+BL/2+tWTR_S = 16 --------
>|
bus:     ............................... p  D0-
D7 ............................ p  D0-D7

clk:    42   43   44   45   46   47   48   49   50   51   52   53   54
55   56   57   58   59   60   61   62
CMD:     .    .    .    .    .    .    .    .    .    .    .    .    .
.   RD   .    .    .    .    .    .
                              |<-------- WL+BL/2+tWTR_L = 18 --------
>|
bus:     ................................................................ p
D0-D7
```

12.3 The turnaround rules

Transition	Group relation	Command spacing	Why
RD -> RD	same	tCCD_L = 6	Long CAS-to-CAS spacing inside one bank group
RD -> RD	different	tCCD_S = 4	Short CAS-to-CAS spacing across bank groups
RD -> WR	either	tRTW = 6 (book symbol)	Read burst + postamble must clear before write preamble; JESD79-4D gives CL - CWL + RBL/2 + 1 + tWPRE, which is 6 here
WR -> RD	different	CWL + BL/2 + tWTR_S = 16	tWTR_S measured from first CK edge after last write data; different group relaxes the delay
WR -> RD	same	CWL + BL/2 + tWTR_L = 18	tWTR_L is longer because the read lands in the same group that is still finishing the write

tWTR is internal and invisible on the bus, but the command spacing rule enforces it. CL does not appear in the WR->RD rule: with CWL programmed independently (MR2), the write tail is positioned by WL alone, and RL runs from the RD command itself.

12.4 Scheduling takeaways

- Bank groups are the bandwidth lever: a read to B1 followed by a read to B5 saves 2 clocks over back-to-back reads in BG0 (tCCD_S=4 versus tCCD_L=6). The scheduler should interleave groups.
- tRRD splits the same way: ACT-to-ACT to different groups uses tRRD_S=2; to different banks inside the same group uses tRRD_L=3.
- tFAW still applies: a run of four ACT commands must fit inside a 12-clock window. If any of the requests above had been row misses, their ACTs would count toward that window; here all rows are hits so no ACTs appear.
- Refresh is another hidden consumer of bandwidth. In 1x mode a REF command is required on average every tREFI (7.8 us at normal temperature) and locks the channel for tRFC1; 2x and 4x fine-granularity refresh modes trade more frequent shorter pauses for the same total refresh load.
- Grouping reads with reads and writes with writes (write batching / read prioritization) is still the single biggest bandwidth lever in mixed traffic, but bank-group interleaving is the new lever DDR3 did not have.

12.5 AXI side

The AXI fabric never sees any of this. Read masters see variable RVALID latency; write masters see variable WREADY/BVALID timing. A controller with enough queue depth reorders requests 1-7 (respecting AXI ordering rules per ID) to convert the trace above into group-interleaved bursts and park turnarounds.

Source: JESD79-4D sections 2.8, 4.9, 4.19, 4.22, 4.23, 4.24, 4.25, 4.25.6, 4.26

13 Data Path: Prefetch, Bursts, Strobes, and Data Integrity

13.1 8n prefetch

DDR4 keeps the 8n prefetch introduced with DDR3: every column access moves eight words per DQ group between the array and the IO buffers in one internal transfer. The array therefore runs at one eighth of the pin data rate. BL8 is the natural transfer; BC4 (burst chop) is a BL8 transfer stopped after four words on the bus. Rules such as tWR and tWTR are anchored to the second half of the internal 8-word movement, which is why fixed-BC4 writes pull those timings in by two clocks while OTF-BC4 writes behave like BL8.

13.2 Burst length and burst order

Burst length is set by MR0 A1:A0 and can be fixed BL8, fixed BC4, or OTF via A12/BC_n during each READ or WRITE. A12=1 selects BL8 and A12=0 selects BC4; A12 is not a column address bit in OTF mode.

MR0 A1:A0	A12 (OTF)	Burst
00	1	BL8 (fixed)
01	0/1	BL8 or BC4 selected by A12
10	0	BC4 (fixed)
11	-	reserved

Burst type (MR0 A3) selects sequential or interleaved ordering. The table below shows the data word order inside the burst.

BL	Start A2:A0	Sequential	Interleaved
BC4 (read)	000	0,1,2,3	0,1,2,3
BC4 (read)	001	1,2,3,0	1,0,3,2
BC4 (read)	010	2,3,0,1	2,3,0,1
BC4 (read)	011	3,0,1,2	3,2,1,0
BC4 (read)	100	4,5,6,7	4,5,6,7
BC4 (read)	101	5,6,7,4	5,4,7,6
BC4 (read)	110	6,7,4,5	6,7,4,5
BC4 (read)	111	7,4,5,6	7,6,5,4
BC4 (write)	0,V,V	0,1,2,3	0,1,2,3
BC4 (write)	1,V,V	4,5,6,7	4,5,6,7
BL8 (read/write)	000	0,1,2,3,4,5,6,7	0,1,2,3,4,5,6,7
BL8 (read/write)	001	1,2,3,0,5,6,7,4	1,0,3,2,5,4,7,6
BL8 (read/write)	010	2,3,0,1,6,7,4,5	2,3,0,1,6,7,4,5
BL8 (read/write)	011	3,0,1,2,7,4,5,6	3,2,1,0,7,6,5,4
BL8	100	4,5,6,7,0,1,2,3	4,5,6,7,0,1,2,3

BL	Start A2:A0	Sequential	Interleaved
(read/write)			
BL8 (read/write)	101	5,6,7,4,1,2,3,0	5,4,7,6,1,0,3,2
BL8 (read/write)	110	6,7,4,5,2,3,0,1	6,7,4,5,2,3,0,1
BL8 (read/write)	111	7,4,5,6,3,0,1,2	7,6,5,4,3,2,1,0

When write CRC is enabled, fixed BL8 writes use A2:A1:A0=0:0:0. For BC4 writes with CRC, A2 selects whether the lower or upper four data words are sent first.

13.3 DQS: the source-synchronous strobe

DQ transitions are referenced to DQS_t/DQS_c, not directly to CK.

- Reads: the DRAM drives DQS edge-aligned with DQ. The read preamble is programmable to 1tCK or 2tCK via MR4 A11; the 2tCK mode is intended for DDR4-2400 and above. Minimum tRPRE is 0.9 tCK for 1tCK mode and 1.8 tCK for 2tCK mode. The read postamble is fixed at a nominal 0.5tCK, with tRPST(min) 0.33 tCK. tDQSCK bounds the strobe rising edge to CK, tDQSQ bounds the latest valid DQ transition to DQS, and tQH bounds the earliest invalid transition. tQSH and tQSL describe the high and low pulse widths of the differential strobe.
- Writes: the controller drives DQS and must keep its rising edge within tDQSS of CK (about +/-0.27 tCK). DQ setup and hold against DQS are tDS and tDH. Write preamble is programmable to 1tCK or 2tCK via MR4 A12; choosing 2tCK increments the first CWL set by two clocks and adds one clock to tWTR. Minimum tWPRE is 0.9 tCK in 1tCK mode and 1.8 tCK in 2tCK mode; tWPST(min) is 0.33 tCK.
- Read preamble training is available in MPR mode via MR4 A10 and can be used for read leveling.

13.4 DM and DBI

x8 and x16 devices share one DM_n/DBI_n/TDQS_t pin per byte lane; x4 devices do not support DM or DBI. The pin is active low.

Function	Control	Behavior
DM	MR5 A10=1	On writes, low DM_n masks that byte so the old value is kept.
Write DBI	MR5 A11=1	On writes, low DBI_n tells the DRAM to invert the received DQ byte.
Read DBI	MR5 A12=1	On reads, the DRAM inverts the DQ byte and drives DBI_n low when more than four bits in the byte are zero.
TDQS	MR1 A11=1	x8 only; enables termination reference for the strobe. Mutually exclusive with DM and DBI.

DM and write DBI cannot be enabled at the same time. When both are disabled, the DRAM turns off the receiver on that pin during writes; when read DBI is disabled, the DRAM turns off the driver on that pin during reads.

13.5 Write CRC

DDR4 provides CRC for writes only, not for reads. The polynomial is the ATM-8 HEC, X^8 + X^2 + X + 1. The controller generates the 8-bit checksum and appends it to the burst. When CRC is enabled the write frame is always 10 UI long for both BL8 and BC4. A x8 device uses one 72-bit CRC tree (64 data bits plus the DM/DBI lane or ones if neither is enabled); a x16 device uses two identical 72-bit trees; a x4 device uses a 32-bit tree. CRC mismatches are reported as a pulse on ALERT_n. When both DM and write CRC are enabled, the DRAM checks CRC before committing data and blocks the write on mismatch; timing parameters switch to tWR_CRC_DM, tWTR_S_CRC_DM, and tWTR_L_CRC_DM.

The data bus does not carry conventional parity or ECC; CRC and CA parity are the DDR4 integrity mechanisms. CA parity and the ALERT_n pin are discussed in the ODT file in this chapter.

13.6 Latency arithmetic, collected

Quantity	Formula	Programmed by
RL (read latency)	AL + CL + PL	MR1 (AL), MR0 (CL), MR5 (PL)
WL (write latency)	AL + CWL + PL	MR1 (AL), MR2 (CWL), MR5 (PL)
RD cmd -> data	RL clocks	
WR cmd -> first DQS	WL clocks	
RD->RD or WR->WR, same bank group	max(tCCD_L, BL/2)	
RD->RD or WR->WR, different bank group	max(tCCD_S, BL/2)	
RD -> WR	BL/2 + 2 clocks (book symbol tRTW; not named in JESD79-4D)	

Source: JESD79-4D sections 4.3, 4.11, 4.16, 4.19, 4.20, 4.21

14 ODT - On-Die Termination

14.1 What it is

DDR4 keeps the data-bus termination on the die and adds a third programmable value. MR1 holds the normal termination strength Rtt_Nom, MR2 holds the write termination strength Rtt_WR, and MR5 holds the parked value Rtt_Park. The ODT pin still enables and disables Rtt_Nom cycle by cycle; Rtt_Park applies when ODT is low if it has been enabled. Dynamic ODT switches a rank that is being written from its nominal/parked value to Rtt_WR without a new MRS command. ODT mode is enabled if any of MR1 {A10,A9,A8}, MR2 {A11,A10,A9}, or MR5 {A8,A7,A6} is non-zero.

14.2 Programming

Rtt_Nom is selected by MR1 A10:A9:A8. RZQ is the external 240 ohm reference resistor connected to the ZQ pin.

A10 A9 A8	Rtt_Nom
0 0 0	disabled
0 0 1	RZQ/4 = 60 ohm
0 1 0	RZQ/2 = 120 ohm
0 1 1	RZQ/6 = 40 ohm
1 0 0	RZQ/1 = 240 ohm
1 0 1	RZQ/5 = 48 ohm
1 1 0	RZQ/3 = 80 ohm
1 1 1	RZQ/7 = 34 ohm

Rtt_WR is selected by MR2 A11:A10:A9:

A11 A10 A9	Rtt_WR
0 0 0	dynamic ODT off
0 0 1	RZQ/2 = 120 ohm
0 1 0	RZQ/1 = 240 ohm
0 1 1	Hi-Z
1 0 0	RZQ/3 = 80 ohm
1 0 1	reserved
1 1 0	reserved
1 1 1	reserved

Rtt_Park is selected by MR5 A8:A7:A6:

A8 A7 A6	Rtt_Park
0 0 0	disabled
0 0 1	RZQ/4 = 60 ohm
0 1 0	RZQ/2 = 120 ohm
0 1 1	RZQ/6 = 40 ohm
1 0 0	RZQ/1 = 240 ohm
1 0 1	RZQ/5 = 48 ohm

A8 A7 A6	Rtt_Park
1 1 0	RZQ/3 = 80 ohm
1 1 1	RZQ/7 = 34 ohm

Dynamic ODT requires the DLL to be on and locked and is not available during write leveling or DLL-off mode.

14.3 Synchronous and dynamic ODT timing

In synchronous ODT mode (DLL on and locked), the ODT pin is sampled on rising CK edges. Rtt_Nom turns on DODTLon clocks after ODT is sampled high and turns off DODTLoff clocks after ODT is sampled low.

Symbol	Meaning
DODTLon	ODT high sample -> termination begins to turn on = CWL + AL + PL - 2 (1tCK preamble) or -3 (2tCK preamble)
DODTLoff	ODT low sample -> termination begins to turn off = same as DODTLon
RODTLoff	Read command -> internal ODT turn off = CL + AL + PL - 2 (1tCK) or -3 (2tCK)
RODTLon4	Read command -> Rtt_Park turn on for BC4 = RODTLoff + 4 (1tCK) or +5 (2tCK)
RODTLon8	Read command -> Rtt_Park turn on for BL8 = RODTLoff + 6 (1tCK) or +7 (2tCK)
ODTH4	Minimum ODT high after assertion or BC4 write = 4 tCK (1tCK preamble) or 5 tCK (2tCK preamble)
ODTH8	Minimum ODT high after BL8 write = 6 tCK (1tCK preamble) or 7 tCK (2tCK preamble)

Dynamic ODT adds the following latencies measured from the write command:

Symbol	Meaning
ODTLcnw	Rtt_Nom/Park -> Rtt_WR switch = WL - 2 (1tCK) or WL - 3 (2tCK)
ODTLcwn4	Rtt_WR -> Rtt_Nom/Park for BC4 = ODTLcnw + 4 (CRC off) or +7 (CRC on), plus 1 more for 2tCK preamble
ODTLcwn8	Rtt_WR -> Rtt_Nom/Park for BL8 = ODTLcnw + 6 (CRC off) or +7 (CRC on), plus 1 more for 2tCK preamble
tADC	skew window for the Rtt change, about

Symbol	Meaning
	0.3 to 0.7 tCK depending on speed bin

A READ command overrides the ODT pin: termination is disabled around the read burst so the DRAM can drive DQ without fighting its own termination.

14.4 Power-down and asynchronous ODT

Synchronous ODT applies in active and idle states and in active/precharge power-down. MR5 A5 can deactivate the ODT input buffer during power-down to save power; when it does, Rtt_Nom is not provided during power-down but Rtt_Park remains if enabled. In DLL-off mode ODT operates asynchronously with tAONAS and tAOFAS turn-on/turn-off delays; Rtt_WR dynamic ODT is not available and Rtt_WR must be disabled by MRS.

14.5 ZQ calibration

ZQCL performs a long calibration of output driver and ODT values. The first ZQCL after reset is allowed tZQinit (1024 nCK); later ZQCL commands are allowed tZQoper (512 nCK). ZQCS performs a short periodic calibration allowed tZQCS (128 nCK). During calibration the channel must be quiet, CKE must be high, ODT must be low, all banks must be precharged with tRP met, and all DQ pins must be high-impedance or in Rtt_Park. ZQ calibration may be issued in parallel with DLL lock time when exiting self-refresh, but an explicit command is required after self-refresh exit (earliest tXS/tXS_Abort/tXS_FAST).

14.6 ALERT_n and data integrity

ALERT_n is not part of ODT but shares the same physical alert path. It reports write CRC errors as a pulse of at least six clocks; on detection the DRAM sets MR5 A3 (CRC Error Clear) and MPR page 1 status. ALERT_n also reports CA parity errors; the two error types cannot be distinguished at the pin and require reading the mode registers. CA parity latency and persistent-error reporting are programmed in MR5.

14.7 What DDR4 does NOT have

Stated plainly to avoid back-porting later features:

- No OCD (off-chip driver calibration). ZQ calibration takes its place.
- No on-die ECC and no link-ECC. System-level ECC, if needed, is built above the DRAM with extra devices.
- No VTT termination. DDR4 uses 1.2 V SSTL-class signaling, not POD.
- DBI is present, not absent; it is controlled by MR5 A11 and A12.

Source: JESD79-4D sections 3.5 (MR1/MR2/MR5), 4.12, 4.16, 5.1, 5.2, 5.3, 5.4, 5.5

15 Core Timing Parameters

The bank-state timings: what each parameter constrains, and the command transition it applies to. DDR4 expresses most core timings in clocks per speed bin (nRCD/nRP track the nominal CL of the bin; see the speed-bin tables); ns-to-clock conversion is divide by tCK and round up (RU).

15.1 Row-cycle parameters

Symbol	Definition	Applies between	Value
tRCD	Row-to-column delay	ACT(bank) -> RD/WR(bank)	nCK per speed bin (10 at DDR4-1600 rising to 24 at DDR4-3200)
tRP	Precharge recovery	PRE(bank) -> ACT(bank)	nCK per speed bin, matched to tRCD
tRAS	Row-open window	ACT(bank) -> PRE(bank)	28-52 nCK by speed
tRC	Full row cycle	ACT(bank) -> ACT(bank)	38-76 nCK by speed (= tRAS + tRP)

15.2 Activate-rate parameters (bank-group aware)

The headline change from DDR3: activate spacing and the four-activate window come in short/long pairs keyed to whether the two banks share a bank group.

Symbol	Definition	Applies between	Value
tRRD_S	ACT to ACT, different bank groups	ACT(grp a) -> ACT(grp b)	4 nCK (all bins; x16 parts pay more at high speed)
tRRD_L	ACT to ACT, same bank group	ACT(grp a) -> ACT(grp a)	5-11 nCK by organization and speed
tFAW	Four-activate rolling window	any 4 ACTs within a sliding window	16-48 nCK by organization and speed

The rolling-window rule is unchanged in shape: if RU(tFAW/tCK) = 16 and an ACT lands at clock N, at most three more ACTs may land in clocks N+1 .. N+15. Interleaving ACTs across the two groups keeps every adjacent pair at tRRD_S; a same-group stream pays tRRD_L per pair and runs into tFAW sooner.

15.3 Data-to-precharge parameters

Symbol	Definition	Applies between	Value
tRTP	Internal analog read must finish before the row closes	internal RD(bank) -> PRE(bank)	max(4 nCK, 7.5 ns)
tWR	Write recovery: last write data committed before precharge	last write data(bank) -> PRE(bank)	15 ns; programmed as nWR in MR0
tWR_CRC_DM	Write recovery with both CRC and DM enabled	as tWR	tWR + max(4-5 nCK, 3.75 ns)

Command-spacing formulas (same bank):

Transition	Earliest spacing (clocks)
RD -> PRE	AL + RU(tRTP) - the spec states the external spacing as AL + tRTP, with tRAS also satisfied
WR -> PRE	WL + BL/2 + RU(tWR)
RDA -> ACT	RU(tRP) after the auto-precharge begins (AL + BL/2 + RU(tRTP) after the RDA, delayed to tRAS), and >= tRC
WRA -> ACT	WL + BL/2 + nWR + RU(tRP), and >= tRC

15.4 Why these numbers dominate

tRCD and tRP remain the row-miss penalty. The new lever is grouping: tRRD_L and tCCD_L make same-group traffic strictly more expensive than cross-group traffic, so the scheduler's real job is to spread column and activate traffic across bank groups - flat-bank DDR3 scheduling heuristics (round-robin the banks) silently become same-group-penalized on DDR4 unless the bank-to-group map is part of the policy.

Source: JESD79-4D sections 4.19, 4.22-4.24, 11.1 (Table 154), 13 (AC tables)

16 Turnaround and Burst Timing

Column-command spacing and bus-direction rules, now bank-group aware.

16.1 Latencies and CAS-to-CAS

Symbol	Definition	Applies between	Value
tCCD_S	Column command to column command, DIFFERENT bank groups	RD/WR -> RD/WR (any type, cross-group)	4 nCK (constant across bins)
tCCD_L	Column command to column command, SAME bank group	RD/WR -> RD/WR (any type, same group)	5-8 nCK by speed
RL	Read latency	RD -> first data	AL + CL; CL 10-24, AL 0 / CL-1 / CL-2 (MR1)
WL	Write latency	WR -> first DQS	AL + CWL; CWL 9-20 paired per speed bin

BL8 occupies 4 clocks of DQ. Cross-group same-direction traffic issues a column command every tCCD_S = 4 clocks - exactly BL/2, so interleaving two bank groups saturates the bus. Same-group traffic pays tCCD_L, so a single-group stream cannot fill the bus no matter how many banks it has: the group, not the bank, is the scheduling quantum.

16.2 Direction changes

Note on tRTW: JESD79-4D does not name the read-to-write parameter (the symbol does not appear in the spec). It gives the relation in the write timing material: RBL/2 + 1 tCK + tWPRE plus the latency offset (CL - CWL, or RL - WL when AL is set). This book calls the parameter tRTW, matching the other books.

Symbol	Definition	Applies between	Value
tRTW	Read-to-write turnaround: read burst and postamble must clear before the write preamble (book symbol; relation unnamed in the spec)	RD(any) -> WR(any)	RL - WL + BL/2 + 1 + RU(tWPRE)
tWTR_S	Write-to-read, different bank groups	last write data -> internal RD (cross-group)	max(2 nCK, 2.5 ns): 2-4 nCK by speed
tWTR_L	Write-to-read, same bank group	last write data -> internal RD (same group)	max(4 nCK, 7.5 ns): 6-12 nCK by speed

Command-spacing forms (from the WR command):

Transition	Spacing (clocks)
RD -> WR	RL - WL + BL/2 + 1 + RU(tWPRE)
WR -> RD	WL + BL/2 + RU(tWTR_S) cross-group; WL + BL/2 + RU(tWTR_L) same-group
RD -> RD / WR -> WR	max(tCCD_S, BL/2) cross-group; tCCD_L same-group

Worked example (the Chapter 4 configuration: CL = CWL = 10, AL = 0, so RL = WL = 10; BL8; tWPRE = 1 tCK; tWTR_S = 2, tWTR_L = 4): RD->WR = 10 - 10 + 4 + 1 + 1 = 6 clocks. WR->RD = 10 + 4 + 2 = 16 clocks cross-group (18 same-group). The group split applies to the write tail exactly where it applies to column spacing - same-group writes into reads are the most expensive transition in the book.

Write CRC stretches the frame: with CRC enabled the write carries an extra CRC beat, write recovery and tWTR reference points move to the CRC field (tWR_CRC_DM), and the write latency may gain a clock - see the data-path chapter.

16.3 Strobe-level notes

- Read preamble is programmable (MR4): 1 tCK or 2 tCK - the 2 tCK setting relaxes the DQS/CK alignment budget on loaded boards.
- Postamble and write preamble values are per the AC tables; tWPRE is the value that appears in the tRTW relation above.
- tDQSCK stays DLL-tight (the DLL is mandatory in normal operation); DQS-to-DQ skew tDQSQ and hold tQH are per the speed-bin tables.

Source: JESD79-4D sections 4.19, 4.24, 4.25, 13 (AC tables)

17 Refresh and Self-Refresh Timing

17.1 Refresh parameters: fine granularity

DDR4 keeps refresh all-bank (there is no per-bank refresh command) and adds fine granularity refresh (FGR): the controller picks the refresh mode in MR4 and gets a matching tREFI/tRFC pair.

Mode	Interval (avg)	tRFC min by density
1x (default)	tREFI1 = 7.8 us (3.9 us above 85 C)	160 / 260 / 350 / 450 (or 350 opt) ns at 2 / 4 / 8 / 16 Gb
2x	tREFI2 = tREFI1 / 2	110 / 160 / 260 / 350 (or 260 opt) ns
4x	tREFI4 = tREFI1 / 4	90 / 110 / 160 / 260 (or 160 opt) ns

Rules:

- Fixed 1x, 2x, or 4x mode allows only the matching REF1x/REF2x/REF4x command; on-the-fly variants of 1x/2x and 1x/4x allow mixing.
- Higher granularity buys shorter busy windows at the cost of more commands: at 8 Gb, 1x stalls 350 ns every 7.8 us (~4.5% duty), while 4x stalls 160 ns every 1.95 us - friendlier to tight schedules, and mandatory thinking when the memory is hot (tREFI halves above 85 C in every mode).
- All banks must be precharged (tRP met) before any REF; the tREFI average bounds the long-term rate, and posting/pull-in slack follows the same 9 x tREFI-style windows as earlier generations.

17.2 Self-refresh

Symbol	Definition	Value
tCKE	Minimum CKE pulse, high or low	max(3 nCK, 5 ns)
tCKESR	Minimum CKE low for self-refresh entry to exit	tCKE(min) + 1 nCK
tXS	Exit self-refresh to non-DLL commands	max(tRFC + 10 ns window per bin)
tXSDLL	Exit self-refresh to DLL-requiring commands	tDLLK (see below)
tDLLK	DLL re-lock time	597 / 768 / 1024 nCK by speed, programmed via MR6

DDR4 adds a self-refresh abort option (MR4 A9): instead of the full tXS wait, a fast-exit path returns to normal operation sooner at the risk of abandoning an in-flight internal refresh - the controller must then issue an explicit REF to make up the gap. Exit is otherwise the familiar pattern: stable clock, CKE high, NOP/DES through the window, then tXS (non-DLL) or tXSDLL (reads, DLL re-locking).

17.3 Power-down timings

Symbol	Definition	Value
tXP	Exit power-down (DLL running) to any valid command	max(4 nCK, 6 ns)
tCPDED	Command pass disable after PD entry	4 nCK
tPD	PD entry to exit window	tCKE(min) to 9 x tREFI

Precharge power-down freezes the DLL; commands needing the DLL (reads) wait out the longer DLL-relock exit, and the ODT turns asynchronous while the DLL is frozen (the tAONPD/tAOFPD windows). Active power-down keeps the DLL running for a tXP-class resume. Power-down performs no refresh: the stay is bounded by the refresh schedule, 9 x tREFI at best with REFs posted ahead.

ZQ calibration runs as ZQCL (long, init/after-power events) and ZQCS (short, periodic) - DDR4 drops the DDR3-era ZQreset form.

Source: JESD79-4D sections 4.9, 4.17, 4.26, 11.1 (Table 154), 13 (AC tables)

18 Package and Electrical Notes

A light pass: what matters when connecting a DDR4 device, skipping the full ballout grids (spec sec 2 has them).

18.1 Packages

- FBGA per JEDEC MO-207; x4, x8 and x16 ballouts (an x32 grid exists for specialized parts; the standard density/addressing tables cover x4/x8/x16). Dual-die and stacked packages present multiple ranks behind one channel with per-rank chip select and ODT.
- New pins versus DDR3: ALERT# (error flag out / connectivity-test in), VPP (wordline boost supply), PAR (CA parity input when enabled), the bank-group address pins BG0-BG1, and the ACT# pin so RAS#/CAS#/WE# pins double as address bits (A16/A15/A14) during non-command cycles.

18.2 Supplies and signaling

Rail	Nominal	Feeds
VDD	1.2 V	core
VDDQ	1.2 V	output drivers
VPP	2.5 V	wordline activation boost (internal charge pump assist)

- SSTL-class 1.2 V signaling; VREFCA at the receiver; the CA bus is the heavily loaded net and is where CAL mode and CA parity spend their effort.
- DQ integrity is now multi-planeed: on-die termination per MR, DBI for power/DC balance, write CRC on the data, and CA parity on the command path - see the data-path and test chapters.

18.3 Speed bins

Grade	tCK (ns)	Nominal CL-nRCD-nRP examples
DDR4-1600	1.25	10-10-10 … 12-12-12
DDR4-1866	1.071	12/13/14-14-14
DDR4-2133	0.937	14/15/16-16-16
DDR4-2400	0.833	15/16-16-16 … 18-18-18
DDR4-2666 / 2933 / 3200	0.75 / 0.682 / 0.625	17/18 up to 22-22-22 / 24-24-24

tCCD_L, tWTR_L, tRRD_L, nFAW and tRFC grow with the grade (the clock- domain tables in sec 11.1 and the AC tables in sec 13 carry the per-bin numbers); same-group and same-bank analog floors stay in nanoseconds, so faster bins pay more clocks for them.

18.4 Characterization

Section 11 defines the IDD/IPP/IDDQ measurement conditions, including the Table 154 loop timings that double as a convenient clock-domain summary of the core parameters. Consult it for power delivery sizing; nothing there changes scheduling.

Source: JESD79-4D sections 2, 11.1, 12-13

19 Test and Calibration Features

DDR4 keeps DDR3's write leveling, MPR, and ZQ calibration and adds DQ Vref training, per-DRAM addressability, CAL mode, and CA parity with an ALERT_n feedback pin.

19.1 Write leveling

Fly-by routing makes CK arrive at each DRAM at a different time, so the controller aligns DQS to CK per device.

Enablement and rules:

- Enter by loading MR1 A7 = 1; exit by loading MR1 A7 = 0. All banks must be idle for entry.
- Only DESELECT and MRS commands are legal in leveling mode. The allowed MRS changes are Qoff (MR1 A12) and the exit bit.
- Dynamic ODT is unavailable; set MR2 A[11:9] = 000 before entry. ODT controls only DQS_t/DQS_c termination; DQ termination stays off.
- Other ranks are usually quieted via MR1 A12 = 1.

Flow: assert ODT after tMOD; wait tWLDQSEN, then drive DQS_t low / DQS_c high; after tDQSL and tWLMRD send one DQS edge; the DRAM returns the sampled CK value on all DQs. Adjust DQS delay until the 0 -> 1 transition is found, then exit and wait tMOD.

Symbol	Meaning	Value
tWLMRD	Entry MRS to first leveling DQS edge	40 nCK
tWLDQSEN	ODT assertion to DQS termination ready	25 nCK
tDQSL / tDQSH	Minimum leveling strobe pulse width	-
tWLO	DQS edge to valid DQ feedback	0 to 9.5 ns
tWLOE	Earliest-to-latest DQ feedback spread	0 to 2 ns
tMOD	Exit MRS to next valid command	max(24 nCK, 15 ns)

19.2 Multi-Purpose Register (MPR)

DDR4 has four MPR locations in each of four pages. Page 0 is read/write; pages 1-3 are read-only.

Page	Purpose
0	Training pattern (MPR0=01010101, MPR1=00110011, MPR2=00001111, MPR3=00000000)
1	CA parity error log
2	MRS readout, including current MR6 VrefDQ range/value
3	Vendor use only

Enablement and rules:

- Enter MPR mode with MR3 A2 = 1. All banks idle and tRP satisfied first.
- MR3 A[1:0] selects the page; MR3 A[12:11] selects readout format. Page 0 supports serial, parallel, and staggered; pages 1-3 support only serial.
- Valid commands in MPR mode are MRS, reads, writes, deselect, refresh, and reset. Reads and writes with auto-precharge behave as plain reads/writes. Power-down entry and self-refresh are prohibited.
- The DLL must be locked before issuing MPR reads.

Flow: PREA -> wait tRP -> MRS MR3 A2=1 -> wait tMRD/tMOD -> RD with A[1:0]=00 and BA1:BA0 selecting the location -> data returns after RL = AL + CL (+ PL). Repeat reads for training, wait tMPRR (1 nCK) after the last burst, then MRS MR3 A2=0 and wait tMRD/tMOD. Page 0 reads may use tCCD_S or tCCD_L; pages 1-3 must use tCCD_L.

19.3 ZQ calibration

ZQ calibration trims Ron and ODT against an external 240 ohm resistor.

Symbol	Definition	Value
tZQinit	First ZQCL after reset	1024 nCK
tZQoper	Later full ZQCL	512 nCK
tZQCS	Short periodic calibration	128 nCK

Issue ZQCL or ZQCS only when all banks are precharged and tRP is met. Keep CKE high and ODT low during the calibration window. ZQCL is required at initialization; ZQCS is used periodically. After leaving self-refresh, wait at least tXS (or the fast/abort equivalent) before issuing a ZQ command. If several DRAMs share one ZQ resistor, their calibration windows must not overlap.

19.4 DQ Vref training

The internal DQ reference voltage is tuned through MR6.

MR6 bit(s)	Function
A7	1 = enable training mode; 0 = disable
A6	Range select (0 = range 1, 1 = range 2)
A5:A0	VrefDQ value

Range 1 spans 60-92% of VDDQ; range 2 spans 45-77%. Step size is 0.5-0.8% of VDDQ (typical 0.65%). After each MR6 value change, wait Vref_time (max 150 ns). Entry delay tVREFDQE and exit delay tVREFDQX are each 150 ns.

Training loop: enter a range, set a value, wait Vref_time, issue dummy writes/reads, and record pass/fail. The current range and value can be read back through MPR page 2. If PDA mode is active, only MRS commands are legal.

19.5 Per-DRAM addressability (PDA)

PDA lets the controller target one DRAM in a rank for an MRS command, useful for different ODT or Vref settings per device.

- Enable with MR3 A4 = 1; exit with MR3 A4 = 0.
- Write leveling must be completed first, and both RTT_PARK and RTT_NOM need to be turned on.
- An MRS is executed only when DQ0 (x4/x8) or DQL0 (x16) is low; when that data bit is high the DRAM ignores the MRS.
- Dynamic ODT is not supported, so only MRS commands may be issued.
- Leave at least AL + CWL + BL/2 - 0.5 nCK + tMRD_PDA + PL between PDA MRS commands.

19.6 CAL mode

CAL delays the command/address bus relative to CS_n so the DRAM can keep CA receivers off between commands.

MR4 A[8:6]	CAL delay
000	Disabled
001	3 nCK
010	4 nCK
011	5 nCK
100	6 nCK
101	8 nCK
110-111	Reserved

tCAL(min) = max(3 nCK, 3.748 ns). In gear-down 2N mode, odd CAL values round up. When CAL is on, tMOD_CAL = tMOD + tCAL and tMRD_CAL = tMOD + tCAL.

19.7 CA parity

CA parity adds an even-parity bit over ACT_n, RAS/A16, CAS/A15, WE/A14, the address bus, bank addresses, bank-group bits, and C0-C2 on 3DS devices. CKE, ODT, and CS_n are not covered.

MR5 A[2:0]	PL	Typical speed bins
000	Disabled	-
001	4 clocks	DDR4-1600/1866/2133
010	5 clocks	DDR4-2400
011	6 clocks	RFU
100	8 clocks	RFU

- MR5 A4 is the parity error status bit; the controller clears it by writing 0.
- MR5 A9 enables persistent parity error mode. In that mode the DRAM resumes parity checking after ALERT_n returns high, even if the status bit has not yet been cleared.
- Parity latency adds to command latency: WL = AL + CWL + PL and RL = AL + CL + PL.
- CA parity is allowed only with DLL on. With parity enabled, only DESELECT may appear between valid commands.

On error, the DRAM ignores the command, logs the frame in MPR page 1, sets the status bit, asserts ALERT_n within tPAR_ALERT_ON, completes in-flight commands, precharges all banks after tRAS_min, holds ALERT_n for at least tPAR_ALERT_PW, then resumes with checking off until the status bit is cleared. Entry/exit delays are tMOD_PAR = tMOD + PL and tMRD_PAR = tMOD + PL.

Write/read CRC is covered in Chapter 5.

19.8 What is not here

- No OCD command set; ZQ calibration replaces it.
- No boundary-scan or JTAG test port inside the DRAM.
- No standard user test mode beyond MPR and vendor-specific MR bits.
- MPR writes are allowed only to page 0; pages 1-3 ignore them.
- No on-die ECC or post-package repair interface exposed to the user.

Source: JESD79-4D sections 4.7, 4.10, 4.10.5, 4.12-4.15, 4.17, 4.17.1